[2] Block Diagram

SCHÉMA BLOC · CHAÎNE DE MESURE (2 CELLULES DE CHARGE) ET ALIMENTATION

HOST / CONNECTORS

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J1 HOST HEADER

JST-XH 7p: +3.3V, GND, SPI

+3.3V

POWER CONVERSION

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TPS61086 BOOST

3.3 V -> 6 V, 1.2 MHz, soft-start

+6V

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LT3042 LDO

6 V -> +5VA, ultralow noise

+5VA

ANALOG SUPPLY

+5VA RAIL

AD7190 AVDD + bridge excitation

+3.3V -> AD7190 DVDD

+5VA

ANALOG FRONT-END

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J4 LOAD CELL 1

5-pin circular: OUT+/-, EXC+/-

OUT+/-

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RC FILTER CH1

R4/R5 + C27/C28/C29

AIN1/2

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AD7190

24-bit sigma-delta ADC
PGA x1...x128, SINC4, chop

AIN1/AIN2 : load cell 1
AIN3/AIN4 : load cell 2
REFIN1 = EXC+/- (ratiometric)
BPDSW : bridge low side
P3 -> excitation switch

P3

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Q1 EXCITATION SWITCH

SI2303 P-MOS, gate = AD7190 P3

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J2 LOAD CELL 2

5-pin circular: OUT+/-, EXC+/-

OUT+/-2

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RC FILTER CH2

R1/R2 + C6/C8/C10

AIN3/4

SPI

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HOST MCU (via J1)

SPI mode 3, SCLK <= 5 MHz

CS, SCLK, DIN, DOUT/RDY, SYNC
(RN1 4x100R series, CS pull-up R8 100k)

WE-SHC shielding frame (H5)

EXC+ (SENSE+, switched by Q1) to both bridges - EXC- returns through AD7190 BPDSW

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Y2 4.9152 MHz

ceramic resonator -> MCLK1/2

LEGEND

- Power rail
- Analog signal
- Digital / SPI
- Clock

SPÉCIFICATIONS CIBLES

Input voltage	+3.3 V (host)
Input current	~46-51 mA (350R bridge)
ADC	24-bit, 2 differential ch.
Reference	ratiometric, EXC = +5VA
Board size	57.0 x 50.75 mm, 4 layers

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